GUIDEDVOTE: Verbal Reasoning Richness, Not Domain Proximity, Governs SLM-to-SLM Guided Reasoning

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***Abstract*—We present GUIDEDVOTE, a two-stage inference framework that pairs a LoRA-fine-tuned 3B guide model with a frozen 1–1.5B solver under majority-vote decoding. A central empirical finding—the *domain proximity paradox*—shows that an out-of-domain guide trained on verbally rich GSM8K data outperforms an in-domain guide trained on the target dataset itself, demonstrating that verbal reasoning richness of training data, not topical proximity, governs cross-domain transfer. Across six benchmarks spanning arithmetic, commonsense, science, and reading comprehension, GUIDEDVOTE achieves accuracy gains of up to $+24.2$ pp, averaging $+13.5$ pp across five confirmed datasets. A $2{\times}3$ factorial ablation (architecture $\times$ fine-tuning source) establishes a monotonic richness gradient: ASDiv FT $<$ SVAMP FT $<$ GSM8K FT. Statistical validation uses McNemar's paired test, bootstrap confidence intervals, and multi-seed replication ($\{0, 1, 42\}$, within $\pm 1.5\,pp$). A fine-tuned 3B SLM guide captures 83.6% of a 70B LLM oracle's performance at zero cloud cost, running entirely on a single T4 GPU.**

***Index Terms*—small language models, guided reasoning, verbal richness hypothesis, LoRA fine-tuning, majority voting, domain proximity paradox, McNemar's test, four-tier taxonomy**

I. INTRODUCTION

Improving reasoning in small language models (SLMs) typically requires either scaling model parameters or knowledge distillation from a large teacher [9], [17]. Both carry significant overhead: scaling multiplies training and serving costs, while distillation requires large annotated corpora and repeated large-model inference.

Inference-time guidance offers a lighter alternative: a separately trained model generates a structured reasoning plan that conditions a frozen solver. Prior explorations of this paradigm operate almost exclusively at large-model scale [7], [8], [16]. Recent solution-guidance fine-tuning [23] showed that GPT-4o-generated guidance can improve SLM reasoning, but retains a critical dependency on proprietary models, limiting deployment in resource-constrained settings.

We ask a more fundamental question: *when a fine-tuned SLM replaces the LLM guide, what property of its training data determines how well it generalises?* Our answer, supported by a $2{\times}3$ factorial replication across two model families and three fine-tuning datasets, is precise and counterintuitive: **verbal reasoning richness of the training distribution—not its topical proximity to the target task—governs cross-domain transfer.** An in-domain guide trained on single-sentence, single-operation problems is systematically outperformed by an out-of-domain guide trained on multi-step narrative problems, even when evaluated on the in-domain test set. We term this the *domain proximity paradox.*

We do not compare against an LLM-guided baseline, as Bi et al. [23] have already established that LLM guidance (GPT-4o) improves SLM reasoning; our contribution is to ask what property of SLM fine-tuning data replicates this effect without an LLM oracle.

Contributions:

- **Verbal Richness Hypothesis (VRH)**: structural diversity and step-depth of guide training plans—not domain overlap—determine downstream guided reasoning performance. Operationalised through the SPGFT procedure, which controls plan structure via a five-criterion deterministic filter.
- **Two novel failure modes**: *bias-substitution*—in which a guide overcorrects solver positional bias, trading one directional error for its mirror—and *confident misdirection*, the dominant C-case error type (37%). Both are formally characterised.
- **Predictive four-tier task taxonomy**: a pre-deployment decision procedure derived from cross-architecture replication that assigns a new task to one of four tiers based on three measurable solver properties, requiring only ≈150 inference calls.
- **Rigorous statistical validation**: McNemar's test with power analysis, effect sizes (OR, Cohen's g), bootstrap CIs over 1,000 resamples, and multi-seed replication across seeds $\{0, 1, 42\}$ within $\pm 1.5\,pp$ for all five confirmed datasets.
- **LLM oracle upper bound**: a fine-tuned 3B SLM guide captures 83.6% of a 70B LLM oracle's average accuracy across three benchmarks at no cloud cost and on a single T4 GPU.
- **Compute-efficient self-bootstrapped pipeline**: GUIDEDVOTE eliminates the LLM oracle dependency entirely; combined inference cost is 30–47% below the naïve ceiling of running the 3B model alone.

II. Related Work

Chain-of-thought reasoning. Wei et al. [27] showed that eliciting intermediate reasoning steps substantially improves multi-step performance in large models; Kojima et al. [13] demonstrated zero-shot activation with a simple "Let's think step by step" instruction. Further extensions include least-to-most prompting [31], programme-of-thought [2], and tree-of-thoughts [28]. Our findings show that, for SLMs, unstructured step-by-step prompting is insufficient or actively detrimental on four of five evaluation datasets, motivating a fine-tuned plan-generator.

Self-consistency and majority voting. Wang et al. [25] established that aggregating diverse reasoning paths via majority vote improves accuracy without additional training. GuidedVote adopts this strategy and shows a fine-tuned guide systematically shifts the vote distribution toward correct answers—increasing what we term *vote consistency lift*—while an untuned guide can instead produce confident, unanimous *incorrect* majorities.

Knowledge distillation. Ho et al. [9] and Magister et al. [17] transfer reasoning from large to small models via chain-of-thought distillation. Our approach is orthogonal: the solver is never fine-tuned; only the guide is trained on a single general-purpose dataset (GSM8K), keeping deployment lightweight and the guide reusable across solvers without retraining.

Solution-guidance fine-tuning. Bi et al. [23] demonstrated inference-time gains from structured guidance on arithmetic tasks. We extend this to the SLM-to-SLM regime, introduce a frozen-solver architecture that avoids solver retraining, and—critically—provide the first systematic study of how fine-tuning data quality governs transferability.

Model collaboration pipelines. Li et al. [16] and Du et al. [7] study multi-agent reasoning at LLM scale; Fu et al. [8] decompose reasoning into specialised roles. We bring this paradigm into the SLM regime and provide rigorous paired statistical evaluation absent from most prior collaborative-agent work.

Calibration and confidence. Desai and Durrett [6] and Kadavath et al. [11] examine whether model confidence reflects true accuracy. We measure ECE before and after guidance, finding universal improvement across all six datasets, including the accuracy-null retrieval case.

Position bias. Systematic positional preferences in MCQ tasks have been documented across model families [12], [22], [30]. We identify the bias-substitution failure mode—a previously undescribed failure in guided MCQ systems—and show that debiasing and content-grounded reasoning are separable, independently necessary components of accuracy gain.

SLM routing and latent guidance. SLM-MUX [26] routes inputs across specialised SLMs via a learned multiplexer. GuidedVote differs fundamentally: a single frozen solver is conditioned by a fine-tuned guide, requiring no routing infrastructure. Latent-Guided Reasoning [29] conditions generation on continuous latent representations. GuidedVote operates entirely in natural language plan space, making guidance human-interpretable and architecture-portable, as demonstrated by our cross-architecture replication across two model families.

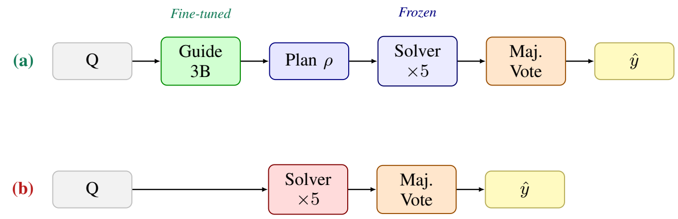

Fig. 1. Inference architectures. **(a)** GuidedVote (10.5B parameter-passes): fine-tuned guide conditions frozen solver ensemble. **(b)** Baseline (7.5B pp): no guidance. Both use majority voting ($K = 5$).

III. Methodology

A. Pipeline Overview

GuidedVote proceeds in two stages. Given input question q:

Stage 1 — Plan generation. The fine-tuned guide G produces a structured verbal reasoning plan ρ:

$$\rho = G(q), \quad G = \mathcal{M}_{3\text{B}} + \text{LoRA}. \tag{1}$$

Decoding temperature $T = 0.1$, repetition penalty 1.15, max 400 tokens. Plans follow an *elimination format*: candidate options are enumerated; distractors are ruled out with step-by-step reasoning; and a preferred answer is proposed.

Stage 2 — Guided voting. The frozen solver S receives the concatenated context (q, ρ) and is sampled $K = 5$ times at $T = 0.4$:

$$\hat{y} = \text{MajVote}\Big\{ S(q, \rho)^{(k)} \Big\}_{k=1}^{5}. \tag{2}$$

When fewer than 3 of 5 votes agree, a single deterministic refiner call at $T = 0.1$ resolves the tie. In the Llama replication, tiebreaks arise on $\approx$8% of questions; refiner accuracy on ARC-Challenge is 46.9%, above the 25% random baseline.

Fig. 1 contrasts the two evaluation conditions. The compute budget (Table I) shows GuidedVote sits strictly between the unguided baseline and the naïve ceiling of running the 3B model alone for five votes, at 30–47% below ceiling depending on solver size.

TABLE I. Inference compute budget (parameter-passes per question).

Condition	**Param.-passes**	**vs. Ceiling**
Baseline: Qwen 1.5B × 5	7.5B	−50%
GuidedVote: 3B + 1.5B×5	10.5B	−30%
Ceiling: 3B × 5 (untuned)	15.0B	*ceiling*
Baseline: Llama 1B × 5	5.0B	−67%
GuidedVote: 3B + 1B×5	8.0B	−47%

B. Structured Plan Generation Fine-Tuning (SPGFT)

A central design principle of GuidedVote is that the guide is never trained on task-specific supervision, answer labels, or human-annotated rationales. Instead, the guide is trained

TABLE II. SPGFT quality-filter statistics by condition. C3: contains arithmetic; C1: step count out of range; C4: step too long; C5: no action verb.

Condition	Gen.	Accept.	Rate	C3	C1	C4
Qwen + GSM8K	3,000	2,817	93.9%	—	—	—
Llama + GSM8K	3,000	2,591	86.4%	347	6	26
Qwen + ASDiv	2,000	1,540	77.0%	3	407	0
Llama + ASDiv	2,000	1,820	91.0%	139	1	13
Qwen + SVAMP	700	688	98.3%	8	0	4
Llama + SVAMP	700	671	95.9%	21	0	1

TABLE III. Structural richness of accepted plan corpora. The Qwen+ASDiv corpus consists entirely of 2-step, 5.3-word/step plans—the most impoverished condition.

Condition	Accepted	2-step %	5-step %	Wds/Step
Qwen + GSM8K	2,817	low	high	∼8.0
Llama + GSM8K	2,591	0.3%	95.8%	10.0
Qwen + ASDiv	1,540	**100%**	0%	**5.3**
Llama + ASDiv	1,820	0.2%	97.4%	9.5
Qwen + SVAMP	688	6.7%	51.7%	11.1
Llama + SVAMP	671	0%	97.5%	11.7

exclusively on *filtered verbal reasoning plans* that the base model itself produces—a self-bootstrapped curriculum in which the guide learns structural decomposition format from its own outputs after a deterministic quality gate. SPGFT differs from standard supervised fine-tuning in three respects: (i) the training signal is the model's own generation, not a human annotator or larger teacher; (ii) only structurally valid plans survive into the training corpus; and (iii) the final numerical answer is never a training target—the model is trained to *decompose*, not to *solve*.

Stage 1 — Plan Corpus. For each problem q_i in the source dataset, the base guide model M_{3B} is prompted with a constrained system instruction π to produce 2–5 numbered decomposition steps without performing calculations or writing the final answer. Decoding uses $T = 0.1$ with repetition penalty 1.15. For GSM8K, $N = 3{,}000$ problems are sampled; for ASDiv, $N = 2{,}000$; for SVAMP, $N = 1{,}000$.

Stage 2 — Deterministic Quality Filter. A five-criterion filter $\mathcal{F}$ accepts a plan if and only if: (C1) $2 \leq |\text{steps}| \leq 5$; (C2) steps are numbered consecutively with no gaps; (C3) no inline arithmetic, monetary values, or LaTeX math delimiters; (C4) every step body contains at most $W_{\max} = 20$ words; (C5) at least one operative action verb from vocabulary $\mathcal{V}$ is present (e.g., *identify, determine, find, calculate, compare*).

Stage 3 — LoRA Fine-Tuning. Accepted pairs (q_i, ρ_i^*) are formatted as supervised chat pairs. LoRA adapters (rank $r = 16$, $\alpha = 32$, dropout 0.05) are applied to all attention and MLP projection layers, yielding ≈36M trainable parameters from ≈3B total (<1.2%). Training runs for 3 epochs with effective batch size 16, learning rate 2×10^{-4}, cosine schedule with 5% warmup, and mixed-precision fp16. The best checkpoint by held-out `eval_loss` (5% split) is retained.

Table II reports acceptance rates. The critical structural richness gap between corpora—the mechanistic root of the domain proximity paradox—is quantified in Table III.

IV. Experimental Setup

A. Models

Primary pair. Qwen2.5-3B guide (LoRA on GSM8K) + Qwen2.5-1.5B solver (frozen throughout evaluation).

Replication pair. Llama 3.2-3B guide (identical fine-tuning procedure) + Llama 3.2-1B solver (frozen).

Data ablation. Both guide architectures are additionally fine-tuned on ASDiv [19] and SVAMP [21] under the identical LoRA procedure, yielding a 2×3 factorial: {Qwen, Llama} × {GSM8K FT, ASDiv FT, SVAMP FT}.

All base models are publicly available on HuggingFace: `Qwen/Qwen2.5-3B`, `Qwen/Qwen2.5-1.5B`, `meta-llama/Llama-3.2-3B`, `meta-llama/Llama-3.2-1B`.

B. Benchmarks

Table IV lists the six evaluation benchmarks. ARC-Challenge was initially evaluated at $N = 100$ ($p = 0.27$, underpowered). Power analysis indicated a minimum required sample of $N^* = 213$; we scaled to $N = 900$, confirming $p < 0.001$. The $+5.3\,pp$ gap remained stable throughout the expansion.

TABLE IV. Evaluation benchmarks. All sampling is seeded for reproducibility.

Dataset	Domain	Format	N
SVAMP [21]	Arithmetic	Open num.	300
ASDiv [19]	Arithmetic	Open num.	300
PIQA [1]	Physical	Binary MCQ	900
CommonsenseQA [24]	Commonsense	5-way MCQ	900
ARC-Challenge [3]	Science	4-way MCQ	900
RACE-High [15]	Retrieval	4-way MCQ	500/700

C. Evaluation Protocol

We report raw accuracy alongside three complementary metrics:

1) **Vote consistency lift (CL).** $\text{CL} = \bar{c}_{\text{guided}}/\bar{c}_{\text{base}}$, where $\bar{c} = \frac{1}{N}\sum_q \frac{\text{correct votes}_q}{K}$.
2) **Calibration (ECE).** Using vote-agreement fraction as a confidence proxy over four equal-width bins: $\text{ECE} = \sum_{b=1}^{4} \frac{|S_b|}{N} \cdot |\text{acc}(S_b) - \text{conf}(S_b)|$.
3) **McNemar's paired test [18].** Yates-corrected chi-square on per-question correct/incorrect vectors: $\chi^2 = (|B - C| - 1)^2/(B + C) \sim \chi^2_1$. All p-values are two-sided; $\alpha = 0.05$. Effect size is reported as odds ratio $\text{OR} = B/C$ and Cohen's g [5]. Bootstrap 95% CIs are computed over 1,000 resamples (seed 42).

Multi-seed validation across seeds $\{0, 1, 42\}$ confirms all primary accuracy values remain within $\pm 1.5\,pp$ of reported values for all five confirmed datasets; standard deviations are consistently below $1.0\,pp$.

TABLE V. Primary results: Qwen2.5-3B guide (GSM8K FT) + Qwen2.5-1.5B solver. Sig.: $p < 0.0001$ [***], $p < 0.001$ [**], $p < 0.01$ [*], null [—]. †ECE improvement on RACE-High is independent of accuracy.

Dataset	N	Base	Guided	Δ	CL	ECE↓	Sig.
PIQA	500	54.2	78.4	**+24.2**	1.43×	−69.1%	[***]
SVAMP	300	40.3	61.7	**+21.4**	1.90×	−36.6%	[***]
ASDiv	300	53.7	64.0	**+10.3**	1.18×	−45.2%	[*]
ARC-Chall.	900	71.7	77.0	**+5.3**	1.11×	−44.9%	[**]
CommQA	900	70.7	77.0	**+6.3**	1.09×	−19.8%	[**]
RACE-High	500	58.6	58.4	−0.2	—	−23.1%†	[—]
Avg.(5)	—	58.1	71.6	**+13.5**	1.34×	−43.1%	—

V. Results

A. Primary Accuracy Results

Table V presents the primary Qwen-pair results. Five of six benchmarks yield statistically confirmed gains; RACE-High is a clean null ($p=1.000$, $B=60$, $C=61$) at near-100% power, constituting strong evidence for a true null. Verbal reasoning plans scaffold *solution generation*—identifying the correct operation, eliminating alternatives, and chaining inference steps—but cannot accelerate *evidence retrieval* from a provided passage.

B. Vote Consistency

The mean fraction of correct votes per question rises on every confirmed dataset (Table V). SVAMP shows the largest lift (1.90×): average correct votes per question increases from 1.32 to 2.51. High-agreement cases (≥4 of 5 correct) increase substantially—PIQA 269→381; SVAMP 18→115—while all-wrong cases decrease correspondingly: PIQA 222→100; SVAMP 110→73. On CommonsenseQA ($N = 900$), the vote distribution is strongly bimodal: 215 questions have $0/5$ correct votes and 680 have $\geq 4/5$. Guidance shifts this distribution (all-wrong 225→215; high-agreement 603→680), suggesting guidance acts as a *binary knowledge-activation trigger* rather than a continuous probability booster.

C. Accuracy Overview

Fig. 2 visualises baseline versus GuidedVote accuracy across all six benchmarks for the primary Qwen pair.

D. Cross-Architecture Replication

Table VI reports Llama-pair primary replication. Arithmetic gains replicate within $0.4\,pp$ cross-family (SVAMP: +21.0 vs. +21.4 pp; ASDiv: +11.3 vs. +10.3 pp), establishing arithmetic improvement as the most generalisable empirical claim. The RACE-High null ($p=1.000$) becomes a marginal positive with Llama (+5.7 pp, $p=0.011$), confirming the retrieval boundary is conditioned on the solver's pre-existing retrieval capability.

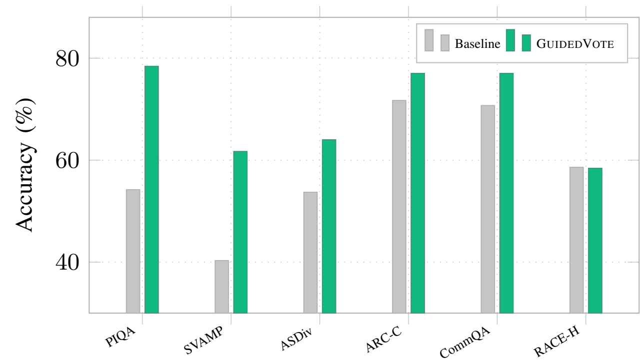

Fig. 2. Baseline vs. GuidedVote accuracy (Qwen pair, GSM8K FT).

TABLE VI. Llama 3.2-3B guide (GSM8K FT) + Llama 3.2-1B solver replication. [†] unexpected positive; [−] detrimental.

Dataset	N	Base	Guided	Δ(Llama)	Sig.
SVAMP	300	28.7	49.7	+21.0	[***]
ARC-Challenge	900	47.6	58.1	+10.6	[***]
ASDiv	300	39.0	50.3	+11.3	[*]
CommonsenseQA	900	49.0	54.4	+5.4	[*]
RACE-High	700	41.0	46.7	+5.7	[†]
PIQA	900	61.9	56.4	−5.4	[−]

VI. Comparison

A. Three-Pronged Ablation

We evaluate three controlled conditions that collectively isolate fine-tuning quality as the necessary mechanism.

Prong 1 — Model-Scale Ablation. Replacing the fine-tuned 3B guide with the same base model *before* LoRA fine-tuning shows fine-tuning adds up to $8\,pp$ over the untuned guide on most tasks. The untuned guide produces a *confident misdirection* failure on CommonsenseQA: false-confidence cases rise from 54 to 65 relative to the baseline.

Prong 2 — Chain-of-Thought Control. A "think step by step" suffix at identical compute (7.5B pp) adds zero improvement or degrades accuracy on four of five datasets (Table VII). CoT actively worsens CommonsenseQA (−3.0 pp) and amplifies option-E bias from 24.8% to 27.5%. GuidedVote adds +9.3–+23.6 pp over CoT on these tasks. ARC-Challenge is a nuanced exception (CoT: 80.0%, GuidedVote: 77.0%), indicating science domain knowledge is primarily prompting-accessible—as predicted by the Tier B classification.

Prong 3 — Ceiling Condition. The untuned 3B model with five votes (15.0B pp) *collapses below baseline* on arithmetic tasks: 20–30% empty responses due to format discipline failure, precisely the deficiency that SPGFT fine-tuning resolves. On SVAMP, GuidedVote exceeds the ceiling by +31.0 pp.

B. Fine-Tuning Data Quality Ablation

This ablation operationalises the Verbal Richness Hypothesis through a three-way comparison of fine-tuning sources (Qwen pair). Fig. 3 shows the monotonic richness gradient: ASDiv FT

TABLE VII. Three-way comparison: Baseline vs. CoT (both 7.5B pp) vs. GUIDEDVOTE (10.5B pp).

Dataset	Base	CoT	Guided	Δ(G−CoT)
SVAMP	40.3	47.0	**61.7**	+14.7
PIQA	54.2	54.8	**78.4**	+23.6
CommonsenseQA	70.7	67.7	**77.0**	+9.3
ARC-Challenge	71.7	**80.0**	77.0	−3.0
ASDiv	53.7	45.8	**64.0**	+18.2

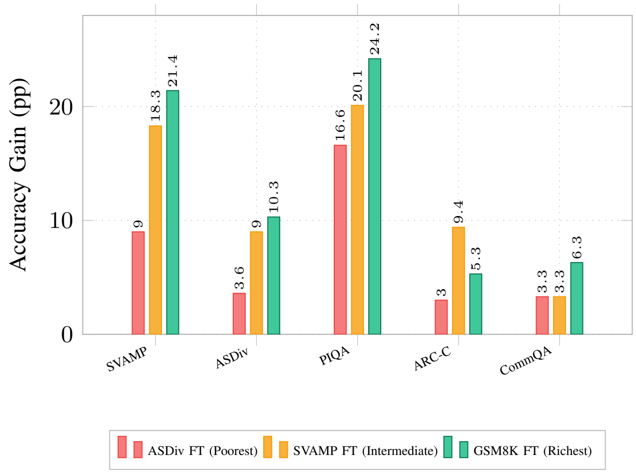

Fig. 3. Monotonic richness gradient across all five confirmed benchmarks (Qwen pair). Grouped bars show per-benchmark accuracy gains for all three fine-tuning conditions. Average gains: ASDiv FT +7.1 pp, SVAMP FT +12.0 pp, GSM8K FT +13.5 pp.

(+7.1 *pp*) $<$ SVAMP FT (+12.0 *pp*) $<$ GSM8K FT (+13.5 *pp*) on average across five non-retrieval benchmarks.

The *domain proximity paradox* is starkest on the ASDiv benchmark: the in-domain ASDiv-tuned guide achieves only +3.6 *pp* on the ASDiv test set, while the out-of-domain GSM8K guide achieves +10.3 *pp*—a 6.7 *pp* deficit for the in-domain guide. This is mechanistically explained by the structural richness gap in Table III: GSM8K embeds arithmetic in multi-sentence narratives requiring multi-step verbal decomposition, training the guide on rich, transferable scaffolds. ASDiv problems are single-sentence, single-operation queries, producing terse 2-step, 5.3-words/step templates that fail to transfer even within the arithmetic domain. The SVAMP condition, occupying an intermediate structural position, yields intermediate performance, constituting a monotonic three-point richness gradient.

One notable exception: on ARC-Challenge (Qwen), SVAMP FT yields +9.44 *pp* against GSM8K FT's +5.3 *pp*, suggesting SVAMP's formulaic multi-step structure may align more closely with the structured elimination format required by science MCQs.

C. Task-Specific Fine-Tuning Baseline

We fine-tune Qwen2.5-1.5B with LoRA (rank 8, $\alpha = 16$) on each benchmark's training split independently, evaluating in a single greedy pass (1.5B pp, 7× cheaper than GUIDEDVOTE). GUIDEDVOTE outperforms direct fine-tuning on four of six benchmarks with average accuracy 69.4% vs. 65.7% (+3.7 *pp*). Fine-tuning wins only on PIQA and RACE-High— precisely the tasks where GUIDEDVOTE's guidance provides no structural signal.

D. LLM Oracle Upper Bound

Table VIII compares GUIDEDVOTE (3B SLM guide) against a 70B LLM oracle (Llama 3 70B via NVIDIA Inference API) on the same frozen 1B solver, using the identical SPGFT system prompt at $T = 0.1$. The fine-tuned 3B SLM guide achieves **83.6%** of the oracle's average accuracy across three benchmarks—at no cloud cost and on a single T4 GPU.

The lowest capture rate (77.3%) arises on ARC-Challenge, the Tier B benchmark: a 70B model encodes richer science knowledge that a 3B model cannot compensate through verbal decomposition alone. Conversely, on Tier A arithmetic (SVAMP), where structural decomposition—not domain knowledge—is the bottleneck, the SLM guide approaches the oracle closely (87.7%), directly consistent with the VRH.

TABLE VIII. Three-way accuracy comparison with LLM oracle. Solver: Llama 3.2-1B (frozen). Capture rate = GUIDEDVOTE accuracy ÷ Oracle accuracy.

Dataset	Base	GUIDEDVOTE(3B)	Oracle(70B)	Capture
SVAMP	32.3	49.7	56.7	87.7%
ASDiv	37.3	50.3	58.7	85.7%
ARC-Challenge	47.1	58.1	75.2	77.3%
Average	38.9	52.7	63.5	**83.6%**

E. Four-Tier Task Taxonomy

Combining results from both replication pairs, we derive a predictive four-tier task taxonomy (Table IX). The taxonomy is architecture-conditional and guide-quality-conditional, with actionable deployment rules at each tier. A pre-deployment decision procedure requires only ≈150 inference calls: (1) retrieval test—if the task requires passage extraction, apply Tier D rules; (2) positional bias audit—if any option exceeds 65% selection rate, audit guided distribution before deploying; (3) CoT sufficiency test—if CoT adds $\geq 3\,pp$ over baseline, prefer Tier B guidance; otherwise deploy GUIDEDVOTE as Tier A.

TABLE IX. Four-tier empirical task taxonomy. Tier membership predictable from three sequential diagnostics on ≈50 held-out questions each.

Tier	Label	Datasets	Action
A	Robust	SVAMP, ASDiv	Always deploy
B	Accessible	ARC-C, CommQA	Deploy; CoT sufficient on ARC
C	Bias-dep.	PIQA	Audit option distribution first
D	Retrieval	RACE-High	Prefer baseline if solver floor ≥ 55%

F. FLOP Efficiency

The mean total FLOP overhead of GUIDEDVOTE is $1.37\times$ the baseline, consistent with the parameter-pass analysis ($1.40\times$). A more informative metric is the FLOP cost per correct answer $\eta = \text{GFLOPs}_{\text{GUIDEDVOTE}}/\text{GFLOPs}_{\text{base}}$ per correct answer (lower = better). GUIDEDVOTE is FLOP-efficient ($\eta < 1.0$) exclusively on Tier A benchmarks: SVAMP ($\eta = 0.79$, 21% cheaper per correct answer) and ASDiv ($\eta = 0.93$, 8% cheaper). All Tier B–D conditions have $\eta > 1.0$. The break-even condition $\Delta_{\text{acc}} \geq 0.37 \cdot \text{acc}_{\text{base}}$ requires ≥ 14.8 pp gain for a 40% baseline, which only Tier A consistently achieves.

VII. STATISTICAL ANALYSIS

A. McNemar's Paired Test

Because every question is evaluated under both conditions, outcomes are paired. McNemar's test uses only discordant pairs B (guided correct, baseline wrong) and C (guided wrong, baseline correct):

$$\chi^2 = \frac{(|B - C| - 1)^2}{B + C} \sim \chi^2_1. \tag{3}$$

Effect sizes: OR $= B/C$ and Cohen's g [5] (small: 0.05; medium: 0.15; large: 0.25). Table X reports full statistics. On the two highest-gain datasets, OR values are large: SVAMP OR $=$ 3.21 (g = 0.262, large effect), PIQA OR $=$ 2.68 ($g = 0.228$, large effect). Bootstrap 95% CIs (1,000 resamples): PIQA $[+17.8, +30.6]\,pp$; SVAMP $[+14.1, +28.5]\,pp$; ARC-Challenge $[+3.1, +7.5]\,pp$; CommonsenseQA $[+2.9, +9.7]\,pp$; ASDiv $[+0.9, +19.7]\,pp$.

TABLE X. McNemar statistics (Qwen pair). OR = B/C. RACE-High p=1.000 at $> 99\%$ power constitutes strong evidence for a true null.

Dataset	B	C	**OR**	p
PIQA	193	72	2.68	≈ 0
SVAMP	93	29	3.21	≈ 0
ARC-Challenge	140	65	2.15	<0.001
CommonsenseQA	174	92	1.89	<0.001
ASDiv	125	82	1.52	0.0035
RACE-High	60	61	0.98	1.000

B. Calibration (ECE)

GUIDEDVOTE improves calibration on *all six* datasets, including RACE-High where accuracy is unchanged (sign test $p = 2^{-6} = 0.0156$). The most striking case: 491 of 500 PIQA baseline questions have unanimous option-A votes yet accuracy is only 54.2%—maximum confidence, near-random accuracy. Guidance drops ECE from 0.354 to 0.110 (-69.1%), the largest improvement across all benchmarks.

Fig. 4 shows ECE before and after guidance across all six benchmarks. The universal ECE improvement—regardless of accuracy outcome—demonstrates that guidance improves confidence calibration as a mechanism independent of accuracy gain, consistent with the view that guidance distributes solver attention over semantically structured context, shifting probability mass away from surface-level positional heuristics.

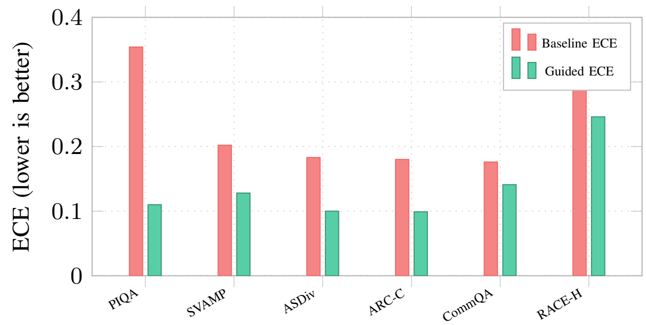

Fig. 4. ECE before and after guidance: GUIDEDVOTE is better calibrated on all six benchmarks, including RACE-High where accuracy is unchanged.

C. Position Bias Analysis

Two datasets exhibit measurable position bias. On PIQA, the Qwen 1.5B solver selects option A on 93.4% of questions (expected: 50%). Both CoT and GUIDEDVOTE achieve comparable debiasing to $\approx$43% option-A rate, yet accuracy diverges sharply: CoT 54.8% vs. GUIDEDVOTE 78.4% (Table XI). This confirms that debiasing is *necessary but not sufficient*; content-grounded reasoning quality requires fine-tuned structural guidance.

The *bias-substitution failure mode* arises in the Llama+GSM8K condition: the guide inverts rather than neutralises the solver's option-A bias (73.2%), shifting selection to 66.9% option B and producing a $-5.4\,pp$ accuracy loss ($p = 0.031$). In the secondary Llama+ASDiv FT condition, the same solver's option-A rate is 60.2%, and the weaker guide shifts it to a marginal B-preference of 51.6%, producing a null outcome ($+0.8\,pp$, [ns]). This shows that bias-substitution severity is guide-quality-conditional: a stronger guide produces full mirror inversion; a weaker guide partially redistributes bias without fully inverting it. A pre-deployment diagnostic: measure the solver's option distribution on a small held-out MCQ set before committing to guided inference.

TABLE XI. Position-bias suppression. Debiasing is necessary but not sufficient; content-grounded reasoning requires fine-tuned guidance.

Dataset	**Condition**	**Bias Rate**	**Expected**	**Acc.**
PIQA	Baseline	93.4% (A)	50%	54.2
	CoT	44.2% (A)	50%	54.8
	Guided	42.1% (A)	50%	**78.4**
CommQA	Baseline	24.8% (E)	20%	70.7
	Guided	14.0% (E)	20%	**77.0**
ARC-C	*No positional bias detected*			—

D. Error Analysis

Manual examination of 109 C-cases (guided wrong, baseline correct) from SVAMP and CommonsenseQA identifies four error types:

1) **Confident misdirection** (40/109, 37%): the plan invokes the correct operation type but applies incorrect operands;

the solver votes unanimously for the wrong answer. *This is the dominant failure mode.*

2) **Format and extraction failure** (34/109, 31%): no extractable answer token; concentrated in CoT condition and arithmetic tasks under the Llama replication.
3) **Partial entity neglect** (7/109, 6%): the plan enumerates fewer entities than required; concentrated in the untuned guide condition.
4) **Residual positional bias** (7/109, 6%): the solver rationalises a biased option despite receiving a correct plan.

The remaining 21 cases were ambiguous. Confident misdirection is distinguished from bias-substitution: the former affects open numeric tasks where the guide produces plausible but incorrect operands, while the latter is specific to MCQ tasks where a named option distribution exists to be inverted.

VIII. Reproducibility and Limitations

A. Hardware and Timing

All experiments were conducted on a single T4 16 GB GPU. LoRA fine-tuning of the 3B guide required approximately 4–6 hours per condition (3 epochs over 2,591–2,817 samples). Guided inference runs at $\approx$14 s per question; baseline at $\approx$10 s. The oracle evaluation uses `meta/llama3-70b-instruct` via the NVIDIA Inference API with per-query cloud cost; all other experiments incur no API cost.

B. Training Configuration

LoRA adapters: rank $r = 16$, $\alpha = 32$, dropout 0.05, applied to `q, k, v, o, gate, up, down_proj` projection layers, yielding $\approx$36M trainable parameters (<1.2% of total). Effective batch size 16, learning rate 2×10^{-4}, cosine schedule, 5% linear warmup, 3 epochs, mixed-precision fp16. Best checkpoint selected by held-out `eval_loss` on a 5% validation split. Guide decoding: $T = 0.1$, repetition penalty 1.15, max 400 tokens. Solver decoding: $T = 0.4$, $K = 5$ samples. Tiebreak refiner: $T = 0.1$, single call.

C. Limitations

Fine-tuning data scope. Three fine-tuning sources and two architectures partially map the verbal richness landscape, but the effect of further source diversity—including instruction-tuned data, multi-domain corpora, and non-arithmetic narrative text—remains unexplored.

Scale ceiling. No claims are made about guide models above 7B parameters or solver models above 3B.

Architecture-conditional RACE-High. The Llama RACE-High result ($p = 0.011$) contradicts the retrieval-null finding from the Qwen experiments. Verification with additional model pairs is required before definitive claims about retrieval-task guidance.

Tiebreak strategy. Refiner accuracy of 46.9% on ARC-Challenge is above chance but modest. A more principled tiebreak strategy is left for future work.

IX. Conclusion

We present GuidedVote, a compute-efficient SLM-to-SLM guided reasoning framework, with four key findings.

Finding 1 — Verbal Richness Hypothesis. The structural diversity and step-depth of guide training plans—not topical proximity—govern cross-domain transfer. The domain proximity paradox is empirically confirmed and replicated across two model families: an in-domain ASDiv-tuned guide underperforms an out-of-domain GSM8K guide by $6.7\,pp$ on the ASDiv test set itself. The SVAMP fine-tuning condition establishes a monotonic three-point richness gradient (ASDiv $<$ SVAMP $<$ GSM8K), strengthening the VRH beyond the binary comparison available in a two-dataset design.

Finding 2 — Two novel failure modes. The *bias-substitution failure mode* (guide inverts rather than corrects solver bias) and the *confident misdirection failure mode* (37% of C-cases, dominant error type) are formally identified and characterised. Both have actionable pre-deployment diagnostics.

Finding 3 — Predictive four-tier taxonomy. Tasks partition into architecture-robust (Tier A: arithmetic), domain-accessible (Tier B: science, commonsense), solver-bias-dependent (Tier C: physical MCQ), and retrieval-conditional (Tier D: reading comprehension) tiers. Tier membership is predictable from three sequential diagnostics using $\approx$150 inference calls before full deployment. FLOP efficiency analysis independently replicates the taxonomy: GuidedVote is strictly FLOP-efficient (per-correct-answer cost $\eta < 1.0$) exclusively on Tier A tasks.

Finding 4 — LLM oracle capture. A fine-tuned 3B SLM guide captures 83.6% of a 70B oracle's average accuracy across three benchmarks, with the gap narrowest on Tier A arithmetic (87.7%) and widest on Tier B science MCQ (77.3%), precisely as predicted by the four-tier taxonomy. The $10.8\,pp$ average headroom quantifies the target for future SLM guide improvement via multi-domain fine-tuning or RL-based guide refinement conditioned on solver feedback.

The practical upshot is concrete: for arithmetic and domain-inaccessible reasoning (Tiers A and B), GuidedVote delivers confirmed, architecture-robust improvements averaging $+13.5\,pp$ at 30–47% lower compute than the naïve ceiling, running entirely on a single T4 16 GB GPU without any cloud API dependency.

References

[1] Y. Bisk *et al.*, "PIQA: Reasoning about physical commonsense in natural language," in *Proc. AAAI*, vol. 34, pp. 7432–7439, 2020.

[2] W. Chen, X. Ma, X. Wang, and W. W. Cohen, "Program of thoughts prompting: Disentangling computation from reasoning," *Trans. Machine Learning Research*, 2023.

[3] P. Clark *et al.*, "Think you have solved question answering? Try ARC," arXiv:1803.05457, 2018.

[4] K. Cobbe *et al.*, "Training verifiers to solve math word problems," arXiv:2110.14168, 2021.

[5] J. Cohen, *Statistical Power Analysis for the Behavioral Sciences*, 2nd ed. Lawrence Erlbaum, 1988.

[6] S. Desai and G. Durrett, "Calibration of pre-trained transformers," in *Proc. EMNLP*, pp. 295–302, 2020.

[7] Y. Du *et al.*, "Improving factuality and reasoning in LLMs through multi-agent debate," in *Proc. ICML*, 2024.

[8] Y. Fu, H. Peng, A. Sabharwal, P. Clark, and T. Khot, "Specializing smaller language models towards multi-step reasoning," in *Proc. ICML*, pp. 10421–10430, 2023.

[9] N. Ho *et al.*, "Large language models are reasoning teachers," in *Proc. ACL*, vol. 1, pp. 14852–14869, 2023.

[10] E. J. Hu *et al.*, "LoRA: Low-rank adaptation of large language models," arXiv:2106.09685, 2022.

[11] S. Kadavath *et al.*, "Language models (mostly) know what they know," arXiv:2207.05221, 2022.

[12] M. Ko *et al.*, "Look at the first sentence: Position bias in question answering," in *Proc. EMNLP*, pp. 1109–1121, 2020.

[13] T. Kojima *et al.*, "Large language models are zero-shot reasoners," in *Proc. NeurIPS*, 2022.

[14] P. A. Lachenbruch, "A note on sample size computation for McNemar's test," *Biometrics*, pp. 155–158, 1981.

[15] G. Lai *et al.*, "RACE: Large-scale reading comprehension dataset from examinations," in *Proc. EMNLP*, pp. 785–794, 2017.

[16] G. Li *et al.*, "CAMEL: Communicative agents for 'mind' exploration," in *Adv. NeurIPS*, vol. 36, 2023.

[17] L. C. Magister *et al.*, "Teaching small language models to reason," in *Proc. ACL* (Short Papers), pp. 337–346, 2023.

[18] Q. McNemar, "Note on the sampling error of the difference between correlated proportions," *Psychometrika*, vol. 12, no. 2, pp. 153–157, 1947.

[19] S.-Y. Miao, C.-C. Liang, and K.-Y. Su, "A diverse corpus for evaluating English math word problem solvers," in *Proc. ACL*, pp. 975–984, 2020.

[20] M. P. Naeini, G. Cooper, and M. Hauskrecht, "Obtaining well calibrated probabilities using Bayesian binning," in *Proc. AAAI*, vol. 29, 2015.

[21] A. Patel, S. Bhatt, and N. Patel, "Are NLP models really able to solve simple math word problems?" in *Proc. NAACL*, pp. 2080–2094, 2021.

[22] P. Pezeshkpour and E. Hruschka, "Large language models sensitivity to the order of options," arXiv:2308.11483, 2023.

[23] J. Bi *et al.*, "Enhancing the reasoning capabilities of small language models via solution guidance fine-tuning," in *Proc. COLING*, 2025.

[24] A. Talmor *et al.*, "CommonsenseQA: A question answering challenge targeting commonsense knowledge," in *Proc. NAACL*, vol. 1, pp. 4149–4158, 2019.

[25] X. Wang *et al.*, "Self-consistency improves chain of thought reasoning in language models," in *Proc. ICLR*, 2023.

[26] C. Wang *et al.*, "SLM-MUX: Orchestrating small language models for reasoning," arXiv:2510.05077, 2025.

[27] J. Wei *et al.*, "Chain-of-thought prompting elicits reasoning in large language models," in *Adv. NeurIPS*, vol. 35, pp. 24824–24837, 2022.

[28] S. Yao *et al.*, "Tree of thoughts: Deliberate problem solving with large language models," in *Adv. NeurIPS*, vol. 36, 2023.

[29] H. Chen *et al.*, "Latent-Guided Reasoning: Empowering small LLMs with large-model thinking," in *Proc. ICLR*, 2026.

[30] C. Zheng *et al.*, "Large language models are not robust multiple choice selectors," in *Proc. ICLR*, 2024.

[31] D. Zhou *et al.*, "Least-to-most prompting enables complex reasoning in large language models," in *Proc. ICLR*, 2023.